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Chapter 1

Introduction

1.1 Motivation

Promptable video trackers such as SAM 3 [5] segment and track any named animal in a video with no training on that species, and conservation teams now deploy them on new species and regions constantly. Yet no rule tells a practitioner, *before running the model*, whether its output is trustworthy for a given animal in a given place.

1.2 Problem statement and research question

This dissertation asks whether SAM 3's zero-shot tracking accuracy on an unseen (species, place) can be *predicted in advance* from properties measurable without any label of the target, and which factors govern that transfer — separately for *finding* the animal and *following* it.

1.3 Hypotheses

H0 (null)
: the label-free distances carry no predictive information about `pDetA`/`pAssA` — transfer is not distance-driven.

H1 (detection)
: `pDetA` falls with species novelty (taxonomic + visual distance).

H2 (association)
: `pAssA` falls with environment difficulty (scene, day/night/IR, clutter, camera motion).

Failing to reject H0 is itself an informative result — and, as Chapter 4 shows, it is what we find: neither a novelty distance nor a follow-up *intrinsic-difficulty* pivot predicts transfer beyond a trivial animal-size effect. Representational probing (does SAM 3 encode taxonomy?) remains the open branch for future work.

1.4 Contributions

To our knowledge this is the first study to *predict*, before the model is run and from label-free distances, the zero-shot transfer of a promptable video tracker — and the first to split that prediction into **detection** (`pDetA`) and **association** (`pAssA`). This distinguishes it from label-free performance prediction for classifiers, which estimates a single scalar from the model's own outputs on the target [2, 9, 15], and from the very recent static-image detector/segmenter estimators [20, 23]; and from camera-trap shift studies, which measure but do not forecast per-site reliability [3, 4, 11]. Concretely we deliver: (i) a rigorous, out-of-sample *negative* result —

under a deliberately powerful test (whole-species and whole-location hold-out with group-cluster-bootstrap significance), label-free before-running distances do *not* predict SAM 3's transfer; (ii) the detection-versus-association decomposition, tested on two near-orthogonal splits — a constructed species hold-out (novelty) and the official location hold-out (environment); and (iii) a *hardened* account of *why* it fails: a follow-up intrinsic-difficulty pivot collapses to the same animal-size confound, so the only label-free property that predicts detection is object size — the very nuisance the analysis controls for — and that positive control confirms the test detects signal when it exists.

1.5 Dissertation outline

Chapter 2 reviews the related work; Chapter 3 describes the data, measurement, distance features, and modelling; Chapter 4 reports the results and out-of-sample validation; Chapter 5 discusses them; Chapter 6 concludes.

Chapter 2

Background and Related Work

This chapter places the dissertation among four bodies of work: promptable segmentation and tracking (the model we study), the tracking metric and its detection/association decomposition (how we score it), label-free prediction of out-of-distribution performance (the statistical idea we extend), and distribution shift in camera-trap vision (the application and the closest empirical neighbours). It closes by stating precisely the gap this work fills.

2.1 Promptable segmentation and tracking

The *Segment Anything* line moved segmentation from a fixed label set to an interactive, promptable formulation, and its successors extended prompting from points and boxes to *concepts*. SAM 3 [5] is a promptable model that, given a short text prompt such as `"impala"`, detects, segments, and tracks every matching instance across a video with no training on that category — a *zero-shot*, open-vocabulary detect–segment–track capability. It is exactly this capability that conservation teams now deploy on new species and regions, and it is the (frozen) model whose transfer we predict. SAM 3 was released alongside SA-FARI [21], a large multi-animal camera-trap tracking dataset that we use as the substrate for our natural experiment (Section 2.6). We treat SAM 3 strictly as a fixed black box: we never fine-tune it, and the only quantity we fit is a small regression *about* its behaviour.

2.2 The tracking metric and its decomposition

Higher Order Tracking Accuracy (HOTA) [13] evaluates multi-object tracking by combining, over a range of localisation thresholds, a *detection* term (`DetA`: were the right objects found?) and an *association* term (`AssA`: were identities kept consistent over time?). SA-FARI scores promptable tracking with a promptable variant, `pHOTA`, that decomposes identically into `pDetA` and `pAssA`. This decomposition is the intellectual core of the dissertation: rather than predict a single scalar, we ask whether *finding* and *following* an animal transfer for *different reasons* — the former with how novel the animal is, the latter with how difficult the scene is — and we therefore model the two components separately throughout.

2.3 Predictable out-of-distribution generalisation

A now-mature literature shows that a model's performance under distribution shift can be anticipated *without* target labels. Miller et al. [15] observe that out-of-distribution (OOD) accuracy is strongly linearly correlated with in-distribution accuracy across many models and shifts, and Baek et al. [2] show that the *agreement* between a pair of models tracks OOD accuracy closely enough to estimate it from unlabelled data alone. A parallel strand estimates accuracy from a